LIN7216 – Formal Linguistics

Plurals and agreement

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Brainstorming

Pronouns, uses and constraints

- What is agreement? Where does it happen in English? In languages different from English? What does it track?
- What is the semantic difference between being singular and being plural, for a DP? Are there some VPs that sound more “plural” than others?

Plan for today

- **Syntactic agreement**, focusing on person/number agreement.
- **Plural ontology**: updating the semantic model.
- **Plural composition**: combining plural subjects and predicates.

Syntactic agreement

Subject-verb agreement: number

(1) a. The girl sing*(**s**).[1] 3.SG

b. The girl**s** sing(***s**). 3.PL

- **Main verbs** agree in **number** (noted #) with their subject.

(2) a. The girl **is**/***are** singing. 3.SG

b. The girl**s** ***is**/**are** singing. 3.PL

- And so do **auxiliaries**, when present.

[1] The *(.) notation means that whatever is inside the parentheses is *not* optional, i.e. *obligatory*.

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Subject-verb agreement: person

(3) a. I **am**/*are/*is lucky/singing. 1.SG
b. You *am/**are**/*is lucky/singing. 2.SG
c. She *am/*are/**is** lucky/singing. 3.SG

- Main verbs and auxiliaries also agree in **person** (noted π) with their subject.
- This is particularly visible with the copula *be*, but also visible with more regular verbs like *sing*.

(4) a. I sing*(**s**) 1.SG
b. She sing*(**s**) 3.SG

Subject-verb agreement: person

(3) a. I **am**/*are/*is lucky/singing. 1.Sg

b. You *am/**are**/*is lucky/singing. 2.Sg

c. She *am/*are/**is** lucky/singing. 3.Sg

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(4) a. I sing*(**s**) 1.Sg

b. She sing*(**s**) 3.Sg

Subject-verb agreement: gender

- In Modern Standard Arabic, verbs often agree in gender with their subject.

(5) a. **ja**-ktub ʔatˤ-tˤa:lib
3.M.Sg-write the-student.M.Sg
'The (male) student writes.'

b. **ta**-ktub ʔatˤ-tˤa:lib-**a**
3.F.Sg-write the-student.F.Sg
'The (female) student writes.'

Subject-verb agreement is “abstract”

(6) a. The sheep bleat(s). b. The fish swim(s).

- Plural agreement on the verb is visible, even when the subject shows **no overt plural marking**.

(7) a. The ox moos. b. The oxen moo.

- It is also there with **irregular plural subjects**.

(8) The albatross soar*(s) over the ocean.

- Plural agreement rightfully disappears when the subject is singular but displays a spurious plural-like ending.
- This all suggests agreement is not a surface phenomenon spotting specific suffixes on the subject and adding corresponding suffixes on the verb. **It manipulates abstract, sometimes covert, features.**

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What are abstract features then?

- Remember how pronouns carry Φ-features: person (π), number (#), and gender. On pronouns, such abstract Φ-features are often quite visible, though sometimes, different feature (combinations) have the same morphological realization:
  2.SG → you
  2.PL → you
- We assume full DPs carry such features too. For instance, *the girls* is 3.PL, *the oxen* is 3.PL, *the albatross* is 3.SG.
- **Abstract features are more systematic/explicit than what morphonology shows on the surface**.
  SG → ∅ / FISH ("SG is unrealized after the word *fish*")
  PL → ∅ / FISH ("PL is unrealized after the word *fish*")
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Agreement as feature valuation

- Subject-verb agreement is a syntactic operation manipulating features, making sure the features of the subject are reflected on the main verb or auxiliary. If proceeds in three main steps:
  - Identification of the subject DP in the tree (**probing**);
  - Copy of the DP's features (**feature valuation**);
  - In many languages, **movement** of the subject to the top of the tree.
- Features copied on the verb may realize in a regular way (think, *sings*), or an irregular way (think suppletive *am*).

3.SG.PRES → -s[2] / sing ("3.SG.PRES is realized as -*s* after the word *sing*")

be → am / [1.SG.PRES] ("*be* realizes as *am* in the presence of [1.SG.PRES][3]")

[2] In fact, the exact surface form is /z/. The fact -s is realized as /z/, /s/, /ɪz/, depending on the ending of the verbal root is called **allomorphy**.

[3] When a root changes in the presence of certain morphological features, we talk about **suppletion**.

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Subject-verb agreement is mediated by T

- Recall matrix sentences are Tense Phrases (TPs), headed by T. T is a functional head hosting tense information and Φ-features.
- Issue: when T gets generated, it is specified for tense but **not Φ-features**! Disregarding gender, we may write: $T \begin{bmatrix} \pi: \\ \#: \end{bmatrix}$
- The missing Φ-feature values are provided by the **subject**, which is specified for them.
- Agreement establishes a formal dependency between the subject and T, which enables the **copy of the subject's features into T's Φ-specification**. For instance:

  $\textsc{Agree}(\text{T}, \text{YOU}_{2.\text{SG}}) \rightsquigarrow T \begin{bmatrix} \pi{:}2 \\ \#{:}\text{SG} \end{bmatrix}$.
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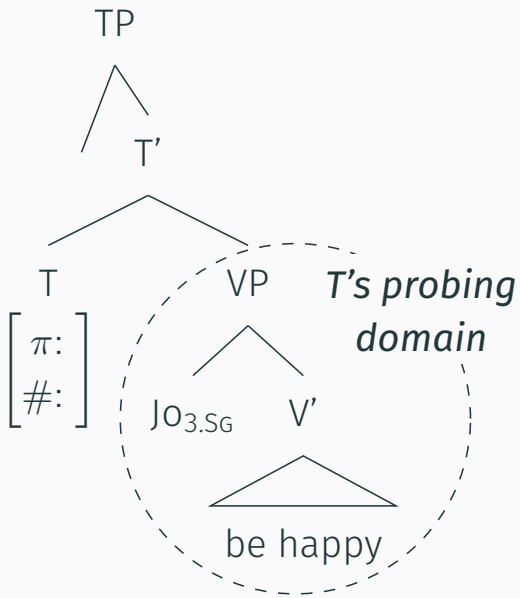
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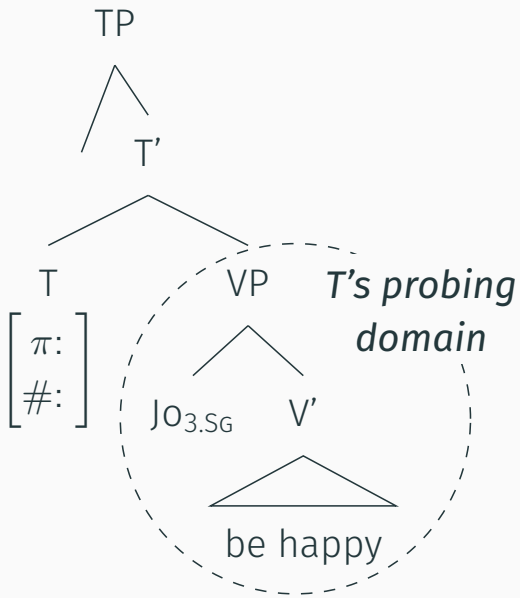
- We assume the subject originates lower than T, in **Spec-VP**.[4]
- In that setup, **T probes downward** for the subject, its **goal**.
- Agreement involves **feature valuation** and **movement** of the subject to Spec-TP (making it sentence-initial).



[4] This might seem weird, but makes other syntactic facts easier to explain, e.g. the VSO word order. It's also quite sensible given that intuitively, the subject is an argument of the verbal complex, and receives θ-role (e.g. AGENT from it). More sophisticated analyses in fact assume that the subject originates in Spec-***v***P, a functional projection slightly above VP.

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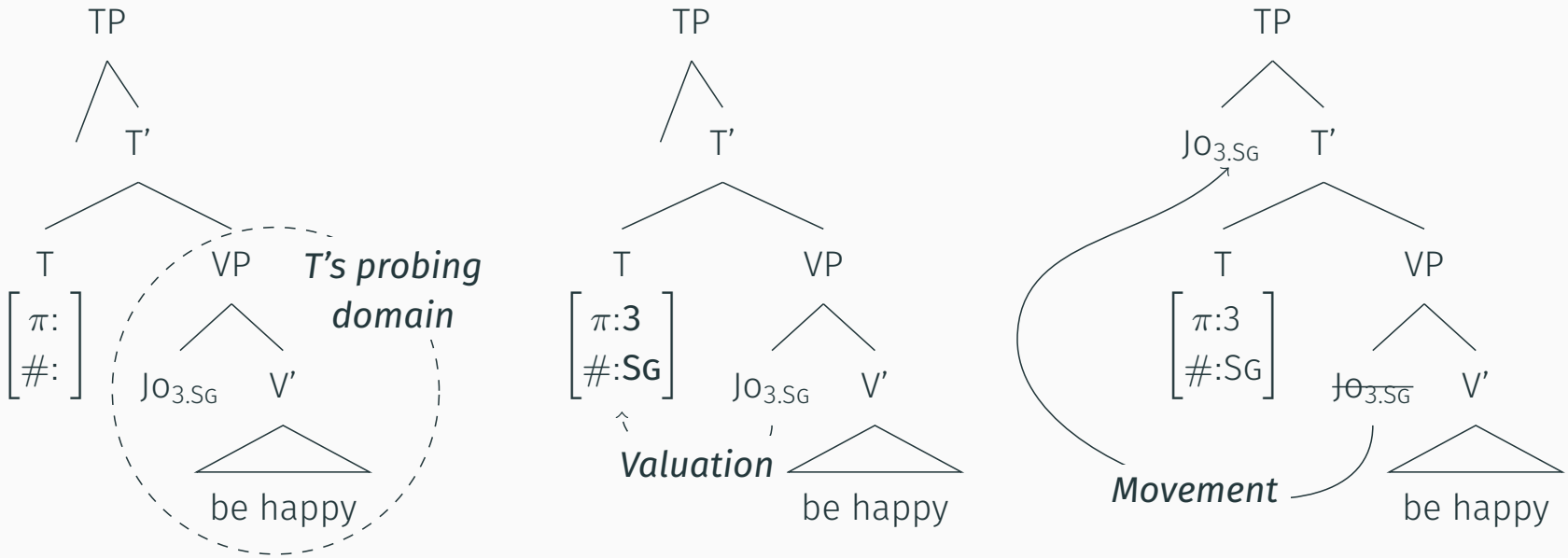
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Bonus: other consequences of subject-verb agreement

- Beyond valuation and movement, subject-verb agreement does two other things: **Case assignment**, and checking of the Extended Projection Principle (**EPP**).
- Case assignment: **T assigns abstract Nominative case to the subject**.[5] In many languages, this case is overtly realized on both pronouns and full DPs. Objects tend to be assigned Accusative *via* a separate Agree process (with a verbal head).

(9) a. **She**$_{3.SG.F.\textbf{Nom}}$ likes **her**$_{3.SG.F.\textbf{Acc}}$

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- EPP: when Agree involves movement of the subject to Spec-TP, an "EPP" feature gets ✓ on T, to mean that its Spec is rightfully filled with something overt (a requirement for many languages).

[5] This does not apply to language with an ergative-absolutive case alignment–Basque, Georgian, Mayan, Tibetan, Sumerian, and certain Indo-European languages such as Pashto, the Kurdish languages and many others.

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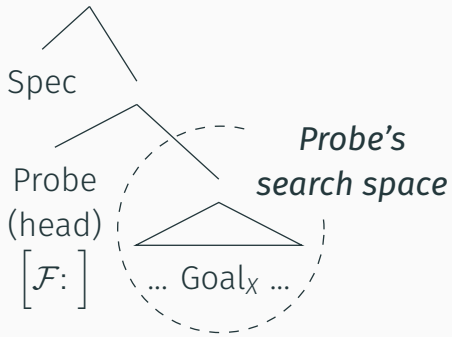
Agreement generalized

- A **probe** is any functional head whose features need to be valued. A **goal** is typically a PP/DP/NP carrying the relevant features.
- The probe searches for a suitable goal downwards, **within its C-Command domain** (i.e. anything that its sister dominates), and stops when it reaches the Spec of the closest CP (**phase edge**).
- Once the probe has found the **closest goal** (in terms of C-Command), it **copies** the goal's features, and may trigger **movement** of the goal to its Spec.[6]

[6] Extra conseuqences may include case assignment (Nom, Acc) and feature checking (EPP, wh...). See previous slide.

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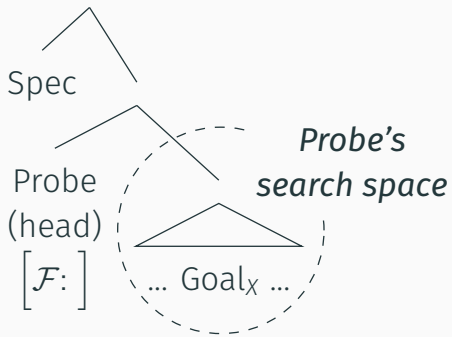
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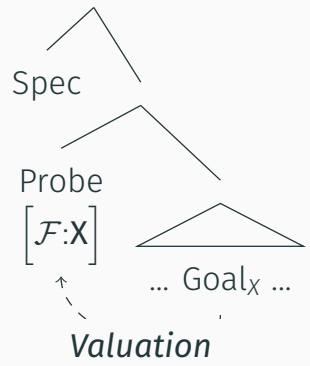


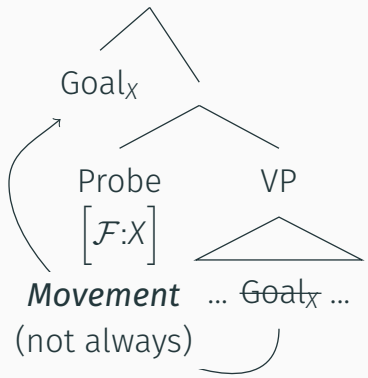
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Agreement recap

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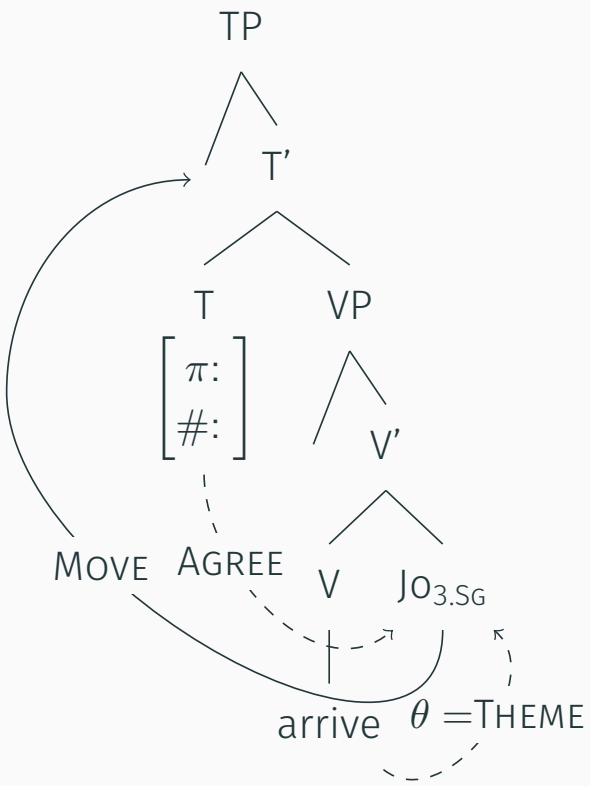
- Agreement establishes a relationship between a **probe** (functional head like T) and an **goal**, usually from the nominal extended projection (PP/DP/NP). Some features of the goal may thus end up on the probe by **feature valuation**.
- Agreement often triggers movement of the goal to the Spec of the probe, establishing a "**Spec-head**" configuration between them.
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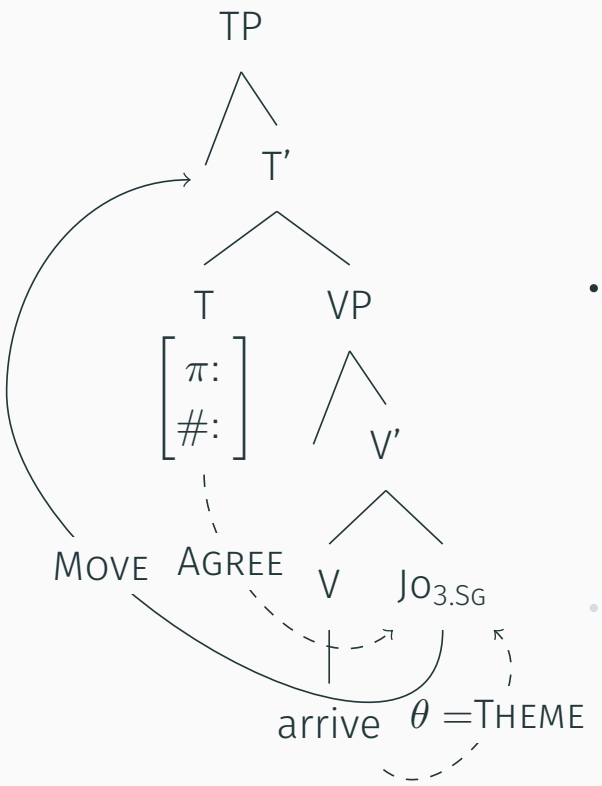
Agreement beyond subject and verb

Unaccusatives



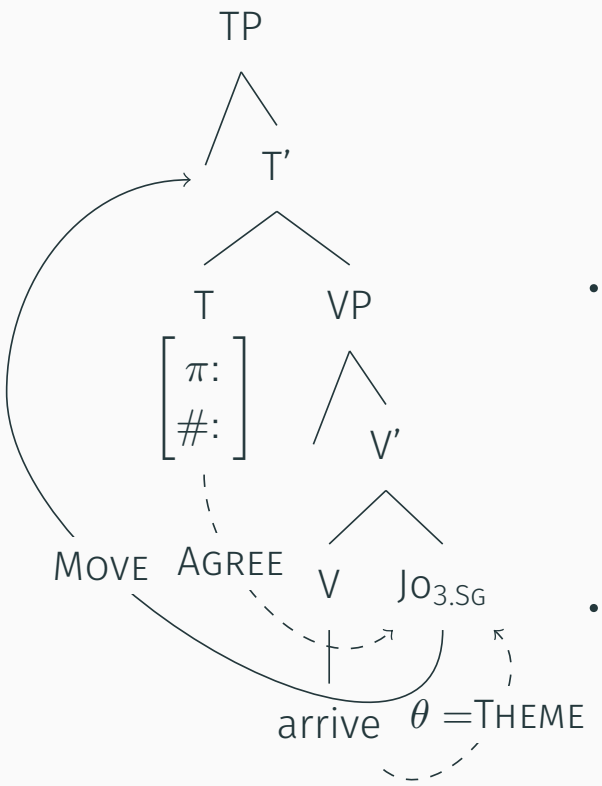
- Unaccusative verbs like *arrive*, *fall*, *die*, *sink* have a **"passive" meaning**: their subject does not perform the action intentionally.
- This is modeled by assuming the surface subject originates in the **object position**, Comp-V, where it gets assigned a Theme θ-role by the verb.
- Because unaccusatives only require an object (they are a subclass of intransitives), it is **selected as closest goal** by the T-probe.
- This triggers feature valuation, and **movement of the object to the surface subject position** (Spec-TP), explaining why unaccusatives end up featuring Theme-subjects!

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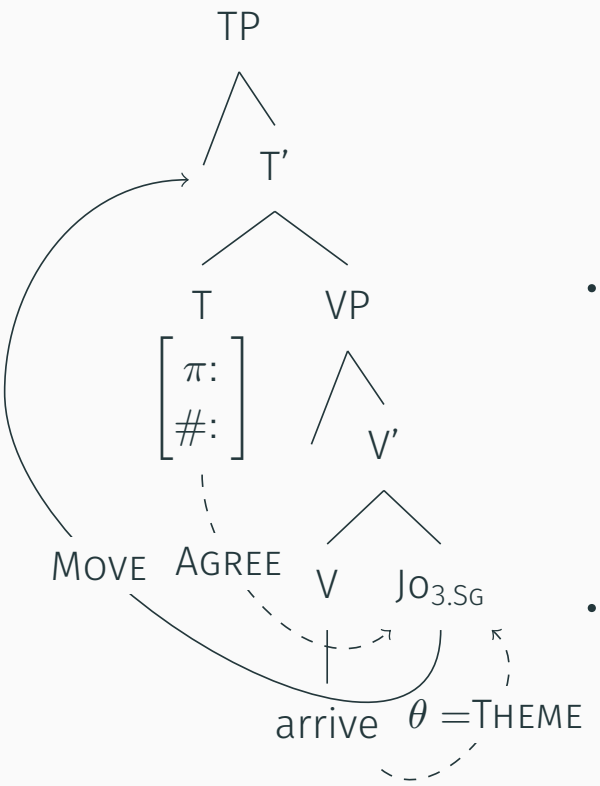
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Bonus: object agreement in Icelandic

(10) a. Henni leidd**u**st þeir
she-DAT was-bored-by-3.PL-st they-NOM
'She was bored with them.'

b. Henni leidd**i**st þeir
she-DAT was-bored-by-3.SG-st they-NOM
'She was bored with them.'

- In Icelandic, certain verbs have "quirky" subjects, which bear an oblique case like Dative (DAT), instead of the expected Nominative (NOM).
- When such verbs are transitive, their object happens to be NOMINATIVE, suggesting **the object agreed with T**!
- Additionally, the main verb optionally realizes the features of the object (in (10a), 3.PL), suggesting once more that T and the object agreed.[7]

[7] In the other case (10b), the verbs displays a default 3.SG form.

Bonus: “symmetric” object agreement in Bantu

(11) Zulu

a. U-mama u-nik-e aba-ntwana in-cwadi.
1a-mama 1SM-give-PFV 2-children 9-book
‘Mama gave the children a book.’ No Object Marking

b. U-mama u-**ba**-nik-e in-cwadi (aba-ntwana).
1a-mama 1SM-**2OM**-give-PFV 9-book **2**-children
‘Mama gave them a book (the children).’ Indirect Object Marking

c. U-mama u-**yi**-nik-e aba-ntwana (in-cwadi).
1a-mama 1SM-**9OM**-give-PFV 2-children **9**-book
‘Mama gave the children it (a book).’ Direct Object Marking

- Many Bantu languages show object marking (OM) in a verbal prefix, which tracks the class of the noun (written in glosses as integers).[8]
- Languages like Zulu are called **symmetric**, because the verb’s OM can track either the features of the Direct Object or those of the Indirect Object.

[8] Noun classes group nouns into categories often delineated along singular/plural and semantic distinctions, like human vs. plant vs. animal.

Bonus: “asymmetric” object agreement in Bantu

(12) Swahili

a. A-li-**m**-pa kitabu.
1SM-PST-**1OM**-give 7.book
‘She gave him a book.’

b. * A-li-**ki**-pa Juma.
1SM-PST-**7OM**-give 1.Juma.
‘She gave it to Juma.’

- Languages like Swahili are called **asymmetric**, because the verb’s OM can only track the features of the Indirect Object (highest goal in the tree)–not those of the Direct Object (lower).

Bonus: omnivorous agreement in Algonquian

- Some languages will agree with any argument, as soon as one of them bears the critical feature (**omnivorous agreement**). In the examples below, the feature is 3(.SG) (third person).

(13) Blackfoot

a. nits-ikákomimm-ok-innaan-**a**
1-love$_{TA}$-INV-1EXC-**SG**
'**She** loves us (exclusive).' Subject agreement

b. nits-ikákomimm-a-nnaan-**a**
1-love$_{TA}$-3OBJ-1EXC-**SG**
'We (exclusive) love **her**.' Object agreement

- This suggests probes can be selective and continue searching deeper, as long as they can't find the target feature.

Bonus: long-distance agreement in Hindi-Urdu

- Long Distance Agreement (**LDA**) happens when a verb agrees with an argument that is not its own, which can happens across clause boundaries.
- This unbounded character is unexpected for that kind of agreement.

(14) Rahul-ne **kitaab** par:h-**ii** th**ii**
Rahul-Erg book.F read-Pfv.F be.Pst.F.Sg
'Rahul had read the book.' Regular object agreement[9]

(15) Vivek-ne [**kitaab** par:h-nii] chaah-**ii**
Vivek-Erg book.F read-Inf.F want-Pfv.F.Sg
'Vivek wanted to read the book.' Long-distance agreement

- Tsez (Nakh-Daghestanian) is also well-known for LDA.

[9] We get object agreement here because the subject is already overtly case-marked.

Bonus: allocutive agreement agreement in Basque

- Most agreement systems target a grammatical entity within the sentence: subject, object etc.
- But in languages such as Souletin, a Basque dialect, agreement can target the addressee. It's called **allocutive agreement**.

(16) a. Pettek lan egin dizü
Peter.ERG worked AUX.3E.ALLOC.POL
'Peter worked.' (formal setting)

b. Pettek lan egin dik
Peter.ERG worked AUX.3E.ALLOC.M
'Peter worked.' (Male addressee in familiar setting)

c. Pettek lan egin din
Peter.ERG worked AUX.3E.ALLOC.F
'Peter worked.' (Female addressee in familiar setting)

d. Pettek lan egin du
Peter.ERG worked AUX.3E
'Peter worked.' (Non-allocutive)

Plural semantics: ontology

What does it mean to be plural?

- We have seen how DP features, including the PL feature, can end up being marked on the verb *via* agreement.
- But what do plurals denote, what kind of verb meanings can combine with them, and what kind of inferences do they allow?

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What does it mean to be plural?

- It's tempting to state that plural DP should denote **sets of individuals**. After all, sets form collections of elements sharing the same type.
- For instance, if there are 3 salients cats, Sachou, Cole and Marm, *the cats* could be thought to denote the set $\{s, c, m\}$.
- This way, we could derive the meaning of *Jo and the cats* by set union: $[\![\text{Jo and the cats}]\!] = [\![\text{Jo}]\!] \cup [\![\text{the cats}]\!] = \{j\} \cup \{s, c, m\} = \{j, s, c, m\}$. So far so good.

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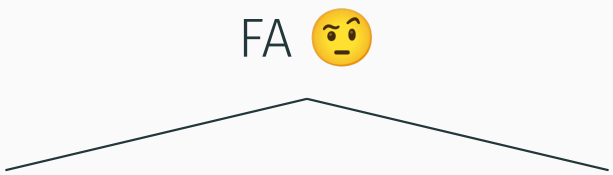
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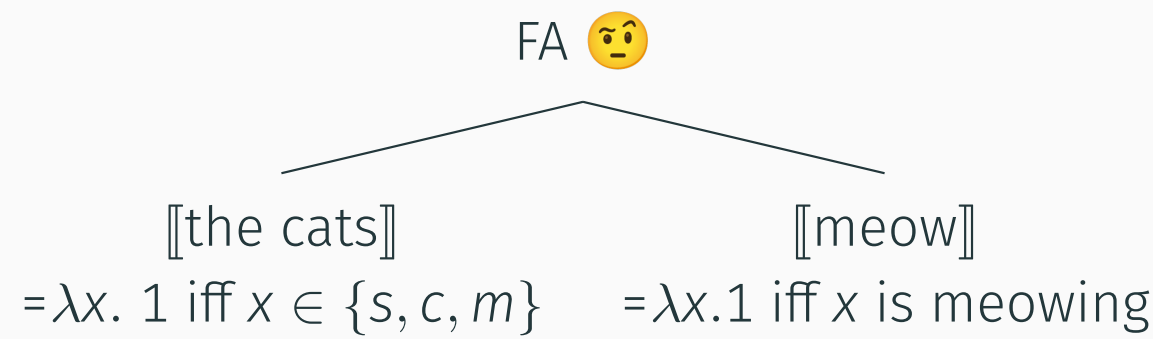


- And at this stage, we want predicate and argument to combine into a truth value-type, so performing set union instead of FA does not make any sense. We have a **composition issue**!

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Summing individuals

- Instead of “lifting” individuals into sets of individuals, we could extend the type of individuals to include **plural individuals**.
- We posit a **sum-formation** operation $\oplus$, distinct from set union, and combining individuals into plural individuals.
- This operation is s.t. $x \oplus x = x$, and its’s also commutative ($x \oplus y = y \oplus x$) and associative ($x \oplus (y \oplus z) = (x \oplus y) \oplus z = x \oplus y \oplus z$).
- We can then state that *the cats* denotes the sum of the salient individual cats (Sachou, Cole and Marm): $s \oplus c \oplus m$. This sum is a plural individual (or **plurality**).
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Generalizing the type of individuals

- Assuming the set of **atomic individuals** (posited on Week 1) is D_e, we can define the set of plural individuals as $D_e^* = D_e \cup \bigoplus_{S \subseteq D_e \wedge |S| \geq 2} S$ ("D_e **plus all the possible sums of individuals forming a subset of** D_e **of size at least 2**"). Note this set includes atomic individuals.
- If D_e were only $\{s, c, m\}$, $D_e^* = \{s, c, m, s \oplus c, s \oplus m, c \oplus m, s \oplus c \oplus m\}$, where $s \oplus m$ for instance, denotes the plurality formed by Sachou and Marm.

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Retrieving the atoms of a plurality

- Given a plural individual $x \in D_e^*$, one can retrieve the set of its atoms using the following recursive Atoms function:

$$\textsc{Atoms}(x) = \begin{cases} \{x\} & \text{if } x \in D_e \quad \text{(base case)} \\ \textsc{Atoms}(y) \cup \textsc{Atoms}(z) & \text{if } x = y \oplus z \quad \text{(inductive case)} \end{cases}$$

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- We can also define what it means to be **part of a plurality**: $x \sqsubseteq y$ iff $x \oplus y = y$. Intuitively, it means y "encompasses" x.
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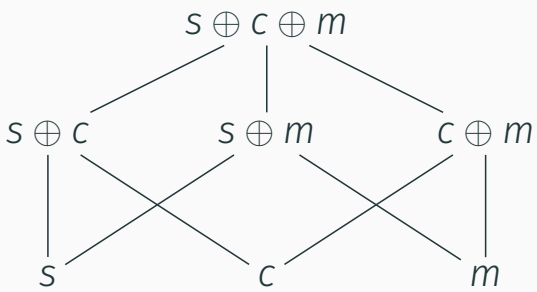
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Representing pluralities

- Pluralities are often represented using **lattices**, graphs in which an ascending line connecting x to y means $x \sqsubset y$ and y is "minimal".[11] If there is an ascending path between x and y, this also means $x \sqsubset y$.
- These lattices can show how pluralities relate to their subparts (**mereological structure**):

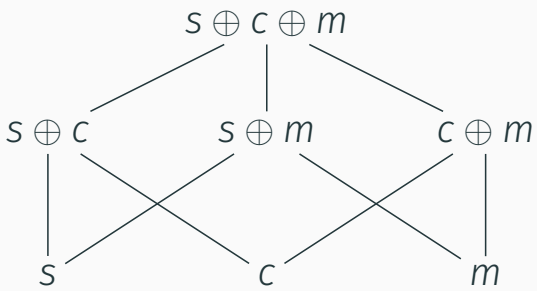


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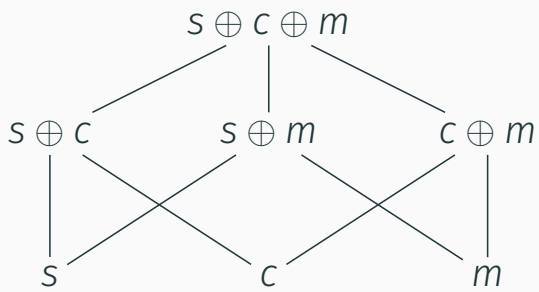


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Representing pluralities

- Pluralities are often represented using **lattices**, graphs in which an ascending line connecting x to y means $x \sqsubset y$ and y is "minimal".[11] If there is an ascending path between x and y, this also means $x \sqsubset y$.
- These lattices can show how pluralities relate to their subparts (**mereological structure**):



- Lattices can also be used to represent the relations between various pluralities collected in a set (i.e., a plural predicate). In that case, the lattice may be asymmetrical/incomplete!

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Plural semantics: composition

Recap and way forward

- We have defined pluralities by extending the type of individuals *via* sum-formation.
- But **how are plural individuals compositionally computed**? For instance, how to derive the denotation of *cats* (plural) from *cat* (singular)?

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Compositional sum formation

- We still assume 3 salient cats, Sachou, Cole and Marm. The singular predicate *cat* denotes the set of individuals that are cats, i.e. $\{s, c, m\} \simeq \lambda x.\ 1$ iff x is a cat.
- We want to derive a denotation for the plural marker -*s*, s.t. the plural predicate *cats* ends up being true of pluralities of cats: $[\![\text{cats}]\!] = \{s, c, m, s \oplus c, s \oplus m, c \oplus m, s \oplus c \oplus m\}$.
- This can be done by assuming -*s* takes a predicates like *cat* as argument and returns **the set of all possible sums of individuals included in it**: $[\![\ \text{-s}\ ]\!] = \lambda P.\ \{\bigoplus S | S \subseteq P\} =$ $\lambda P.\ \lambda x.\ 1$ iff $\exists S \subseteq P.\ x = \bigoplus S$.
- $[\![\ \text{-s}\ ]\!] = \lambda P.\ \{\bigoplus S | S \subseteq P\}$ is often called the $*$ ("**star**") **operator**. Note this operator can in principle be active outside plural morphology.

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How do plurals really mean plural?

- It might seem surprising that the denotation of *cats* **contains atomic individuals** (in our toy example, Sachou, Cole, and Marm).
- This implies that a grammatical sentence like *The cats are meowing* could be understood as *Sachou is meowing*.
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A pragmatic solution to the plural dilemma

- The often strict plurality of plural predicates can be derived pragmatically:
  - Singular predicates **presuppose** their argument is atomic;
  - Plural predicates are **presuppositionless** (argument can be atomic or strictly plural);
  - An independently motivated pragmatic principle called ***Maximize Presuppositions!*** states that if two expressions are true in context, **the one with the strongest presuppositions should be favored**.
- If a speaker wants to talk about an atomic entity then, they can in principle use the singular *or* the plural form, but because the singular has a stronger atomicity presupposition met in context, it should be preferred as per *Maximize Presuppositions!*.
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Types of plural predicates

Distributive vs. collective

- We have just seen how to derive the denotation of plurals from their singular counterpart, *via* the pluralizing star operator $*$.
- What about verbal predicates? Compare the following verbal predicates applying to plural subjects:

(17) a. The cats are meowing. $\rightsquigarrow$ Sachou is meowing.
b. The cats are gathering. $\not\rightsquigarrow$ Sachou is gathering.
c. The cats opened the door. $^{?}\rightsquigarrow$ Sachou opened the door.

- Some predicates like *meow* definitely apply to their individual atoms, some like *gather* definitely don't, and some like *open* allow the inference to hold depending on context.

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Distributive predicates

(17a) The cats are meowing. $\rightsquigarrow$ Sachou is meowing.

- Distributive predicates, like *smile*, *sleep*, *arrive*, *die*, are s.t. **if they hold of a plurality, they hold of its atoms**: if $P(x) = 1$ then $\forall y \in \textsc{Atom}(x).\ P(y)$.
- Let's define a **distributivity operator** D that applies to a plural predicate P and makes sure the above condition holds: $D = \lambda P.\ \lambda x.\ 1$ iff $\forall y \in \textsc{Atom}(x).\ P(y)$.
- Assume *meow* is true of s and c. Then $[\![\ D \text{ meow}\ ]\!] = \lambda x.\ 1$ iff $\forall y \in \textsc{Atom}(x).\ y$ meows = the set of pluralities whose atoms all meow = $\{s, c, s \oplus c\}$.
- Then, $[\![$ the cats D meow $]\!]$ will be true iff $[\![$ the cats $]\!]$ refers to s, c, or $s \oplus c$, and by assumption, all the atoms of these pluralities are meowing! So the distributivity inference holds.

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Collective predicates

(17b) The cats are gathering. $\not\rightarrow$ Sachou is gathering.

- Collective predicates, like *surround*, *meet*, *form a circle*, *cooperate*, are never true of atoms.[12]
- The existence of collective predicates teaches us that some predicate "natively" apply to plurals, and the $*$ operator is not responsible for all the plural readings of predicates.

[12] It's also not true that if they hold of a plurality, they hold of any of its non-atomic subparts. For instance, on can say Sachou, Cole, and Marm met when Sachou met Cole, and Cole met Marm, but Sachou never met Marm.

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Mixed predicates

(17c) The cats opened the door. $^{?} \rightsquigarrow$ Sachou opened the door.

- (17c) can be true in a context where Sachou, Cole and Marm *together* opened the door, but none managed to do so individually. That's a **collective reading**.
- (17c) can also be true in a context where Sachou, Cole, and Marm *each* opened the door at different times, but never did it together. That's a **distributive reading**.
- The availability of both readings is easily derived by assuming the *D* operator is optional with predicates like *open the door*, *lift the piano*, *weigh 10 kg* etc.

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Bonus: weird plural effects

(18) a. The cats meowed. b. The cats didn't meow.

- (18a) often means that *all* the cats meowed, while (18b) means *none* did. This is unexpected given that (18b) is the negation of (18a), and negating a universal statement normally leads to an existential statement (i.e. (18b) should mean *some cats didn't meow*). This all-or-nothing effect is called **homogeneity**.
- Additionally, (18a) allows exceptions, unlike *All the cats meowed*. This is called **non-maximality**.

(19) Cats meow.

- Lastly, the bare plural in (19) enables a **generic reading** of the sentence: *cats typically meow*. This is again exception-tolerant, but in a law-like way, distinct from non-maximal readings.

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